Vaasu Chandra
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Jabalpur, Madhya Pradesh, India

Technical Skills

Programming Languages: C++, Python, Java, JavaScript, SQL, HTML/CSS
AI/ML & Data Science: Machine Learning, NLP, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, XGBoost, Flask
Algorithms & Data Structures: Geometric Algorithms, Mesh Generation, Mesh Manipulation, Graph Algorithms, Object-Oriented Design
Engineering & CAD/CAE: Finite Element Method (FEM), CAE Concepts, Product Lifecycle Management (PLM)
Parallel & Distributed Computing: Apache Kafka, PySpark (Parallel Processing), Docker
Databases & Cloud: MongoDB, MySQL, Cloudinary, Microsoft Azure
Tools & Platforms: Git/GitHub, Postman, Agile/Scrum, MLflow, Apache Airflow
Relevant Coursework: Data Structures & Algorithms (DSA), OOPs, DBMS, Operating Systems (OS), Applied Mathematics

Work Experience

Software Development Intern **Oct 2025 – Dec 2025**
Madhya Pradesh Poorva Kshetra Vidyut Vitaran Company Limited (MPPKVVCL) *Jabalpur, Madhya Pradesh*

- Developed a **Flutter-based mobile application** for Android and iOS using **object-oriented design principles**, delivering real-time electricity consumption insights to **10 Lakh+ consumers** across the distribution network.
- Engineered secure backend integration via **RESTful APIs** with token-based authentication and IVRS-based consumer identification, enabling reliable retrieval and processing of live metering data – demonstrating strong **collaborative team development** practices.
- Implemented **TOD (Time-of-Day)** differential logic to disaggregate raw energy register readings into structured **T1–T4 tariff-wise data structures** for actionable billing insights, following **Agile development** workflows.

Projects

Real-Time Fraud Detection with Streaming ML | *Python, C++, Kafka, PySpark, MLflow* Repo **Apr 2025 – Aug 2025**

- Built an **end-to-end** streaming ML pipeline ingesting 284K+ transactions via **Apache Kafka**, using **PySpark Structured Streaming** and custom **data structures** for real-time XGBoost fraud inference achieving **86.73% recall** and **0.976 ROC-AUC**.
- Applied **SMOTE-based class balancing** on a 0.17% fraud-rate dataset, tracked experiments via **MLflow Model Registry**, and deployed a **FastAPI** inference endpoint containerized with **Docker** and **Kafka + Zookeeper** orchestration.

MLOps Churn Prediction Pipeline | *Python, MLflow, Evidently AI, FastAPI, Docker, Airflow* Repo**Jan 2026 – Mar 2026**

- Built an **end-to-end** MLOps pipeline for customer churn prediction using **Random Forest** with **object-oriented design**, achieving **80.27% accuracy** and **0.86 ROC-AUC** on the IBM Telco dataset (7K+ records).
- Integrated **MLflow** for experiment tracking and **Evidently AI** for automated **data drift detection**; deployed a **FastAPI** REST endpoint and **Airflow DAG** for drift-triggered retraining, containerized via **Docker**.

Geometric Mesh Processing Engine | *C++, Python, OOP, Data Structures* **Personal Project**

- Implemented **geometric and mesh manipulation algorithms** in **C++** using **object-oriented design**, including half-edge data structures, mesh traversal, and surface mesh generation routines – directly relevant to CAE element processing.
- Designed efficient **graph-based data structures** for mesh topology with O(1) adjacency queries and explored **parallel programming** techniques using multi-threading and cache-friendly memory layouts for high-performance computation.

Education

Vellore Institute of Technology **Aug 2022 – Apr 2027**
Integrated M.Tech in Computer Science and Engineering – CGPA: 8.61 *Bhopal, Madhya Pradesh*

Certifications & Achievements

- **Microsoft Azure Fundamentals (AZ-900):** Microsoft
- **IBM AI Engineering:** Coursera
- **Python Data Structures:** Coursera
- **Genesis X Hackathon (First Place):** Secured first place by executing comprehensive data analysis and delivering an end-to-end technical solution under competition constraints, working in a **collaborative team environment**.
- **StockQuest (Winner):** Topped a real-time stock market simulation event, demonstrating strong analytical thinking and data-driven decision-making under dynamic conditions.